

Weekly Internship Report (6/6/26 - 12/6/26)

WEEK 4

Intern's Name: Atharv Gupta

Internship Week: Week 4 (6/6/26 - 12/6/26)

Company/Organization Name: Red Kross Research Foundation

Mentor's Name: Shivam Mittal & Ojas sir

Week Overview:

- Conducted research on OSINT tools including Yadigger, Dedigger, ShortDigger, and NAPALM FTP Indexer.
- Studied public file exposure, data leakage identification, and attack surface discovery techniques.
- Researched the working mechanisms of Multipoint Locks, Tubular Locks, RFID Locks, and Safe Locks, and generated an educational video on the topic.
- Performed phishing URL analysis and threat intelligence investigation.
- Completed Google & OpenDNS Phishing Quiz activities.
- Prepared a research paper and presentation on Predicting Missing Information in Vulnerability Reports.
- Studied Enterprise Security Culture Policy with focus on Governance, Risk, and Compliance (GRC).
- Explored the Metasploitable vulnerable environment for security testing and understanding common vulnerabilities.

Learning Outcomes:

- Improved understanding of OSINT-based reconnaissance techniques.
- Learned methods for identifying publicly exposed files and information leakage.
- Gained knowledge of physical security systems and access control mechanisms.
- Enhanced phishing detection and threat intelligence investigation skills.
- Developed understanding of vulnerability reporting and security research methodologies.
- Learned the fundamentals of Governance, Risk, and Compliance (GRC).
- Gained practical exposure to vulnerable environments using Metasploitable for security learning.

Challenges Faced:

- Understanding different OSINT tools and their practical applications.
- Analyzing phishing indicators and threat intelligence data.
- Researching vulnerability report prediction methodologies and GRC concepts.

Goals for Next Week:

- Study Social Engineering and Human Manipulation Techniques.

- Explore Car Hacking and Automotive Security concepts.
- Understand real-world attack vectors targeting humans and connected vehicles.
- Continue improving cybersecurity research and analytical skills.

Mentor Feedback and Suggestions:

- Continue strengthening cybersecurity research and practical analysis skills.
- Maintain proper documentation of learning activities and findings.
- Focus on understanding both technical and human-centric security threats.

Additional Notes:

- This week provided practical exposure to OSINT, phishing analysis, security research, GRC concepts, and vulnerable testing environments.

Student Diary (Log) Entry

Week	Task Assigned	Activities Performed	Key Learnings	Additional Remarks
1	Cybersecurity Fundamentals and Research Tasks	Malware analysis, GitHub repository setup, Dridex research, Steganography challenge, Cryptography exercises, AI SOC and ZTNA discussions	Learned malware analysis, hashing vs encryption, cryptographic workflows, GitHub documentation, and cybersecurity research concepts	Improved practical understanding of cybersecurity operations and research methodologies
2	Threat Intelligence, Container Security and Web Security Research	Studied MD5 hash extraction, Threat Intelligence concepts, Docker and Docker Networking, Kubernetes basics, Ransomware workflow, OSINT and CSINT, Threat Intelligence Framework, GHDB research, Burp Suite and FoxyProxy, and performed XXE Injection vulnerability analysis using Burp Suite	Learned threat intelligence concepts including CVE, TTPs, Pyramid of Pain, threat actors, OSINT/CSINT methodologies, container technologies, web vulnerability assessment, and XXE exploitation techniques	Enhanced practical cybersecurity research, threat analysis, reconnaissance, and web application security skills
3	Web Application Security Assessment and Penetration Testing	Network Security Assessment, OWASP ZAP security testing, Prototype Pollution labs, Web LLM Attack labs, and web application penetration testing exercises	Learned prototype pollution, DOM XSS, OWASP ZAP usage, Web LLM attack concepts, and penetration testing methodologies	OWASP ZAP usage, Web LLM attack concepts, and penetration testing methodologies Strengthened practical web application security and vulnerability assessment skills
4	OSINT, Threat Intelligence and Security Research	Researched OSINT tools, lock mechanisms, phishing analysis, OpenDNS phishing quiz, Metasploitable environment, vulnerability report research, and GRC concepts	Learned OSINT reconnaissance, data leakage identification, phishing detection, vulnerable environment analysis, and enterprise security governance	Improved analytical and cybersecurity research capabilities

